# Text Representation

1) OHE
2) BOW
3) N-grams
4) TF-IDF

## One Hot Encoding

4 elements

0 [1, 0, 0, 0]
1 [0, 1, 0, 0]
2 [0, 0, 1, 0]
3 [0, 0, 0, 1]

Binary Vectors

Vector length
⇓
Vocabulary Size

| | cat | dog | fish | bird |
|---|---|---|---|---|
| Cat → | 1 | 0 | 0 | 0 |
| Dog → | 0 | 1 | 0 | 0 |
| fish → | 0 | 0 | 1 | 0 |
| bird → | 0 | 0 | 0 | 1 |

## Advantages

Simple & Intuitive

Easy to Implement

No Info loss

## Disadvantages

Huge Vocabulary Size
* high dimensional *

Extremely Sparse vectors

No Semantic meaning

* All words are equi valent distant.

Can't handle out of vocab words

## 2) Bag of Words (BOW)

Text Document Representation

Sen 1 :- John likes to watch movies. Mary likes too.

Sen 2 :- John likes to watch football.

Dict

{ 1 : John , 2 : likes , 3 : to , 4 : watch , 5 : movies

6 : football , 7 : Mary , 8 : too }

## Text Representation (No of Appearances)

{ John : 2, likes : 3, to : 2, watch : 2, movies : 1, football : 1, Mary : 1, too : 1 }

## Vectors (Index)

[1, 2, 3, 4, 5, 7, 2, 8] X

[1, 2, 3, 4, 6] X

Vocabulary Size = 8

[2, 3, 2, 2, 1, 1, 3, 1]

[2, 3, 2, 2, 2, 1, 0, 0, 0]

Vectors (Based on no of appearances)

# Ignoring grammar OR order

## Vocabulary

[cat, dog, bird] {1x3}

Doc 1: cat dog cat → [2, 1, 0]

Doc 2: dog bird → [0, 1, 1]

Doc 3: cat cat cat cat → [4, 0, 0]

Dimension :- 3

## Advantages

1) Simple & Intuitive
2) Captures Word Frequency
3) Can handle vocab size

## Disadvantages

1) Losses order
2) Sparse Vectors
3) Common words dominate more.

{ Dog bite the man → Doc 1
  Man bite the dog → Doc 2 }

Vocab :- [Dog, bite, the, man]

[1, 1, 1, 1]

[1, 1, 1, 1]

Dictionary :- 1 million words

Sen :- 10 words

Vocab Size :- $[1,000,000]^{*}_{-10}$

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## N-GRAMS

GRAM :- A word

N - (number i.e, 1, 2, 3, 4 5 ... etc)

1 - GRAM :- uniquam

2 - GRAM :- biquam

3 - GRAM :- triquam

4 - GRAM :- quadquams